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Computer-based privacy protection for chat groups in a virtual environment

Abstract

In an approach to improve the privacy of chat groups within a virtual world, in response to identifying a private chat group has been created between two or more users, embodiments collect, by an internet of things (IoT) sensor set, a voice for each user in the private chat group. Further, embodiments analyze, by an analysis module, the collected voice, an intonation associated with the collected voice and an original pronunciation associated with the collected voice, and generate a de-content pronunciation based on the analysis of the collected voice, the intonation, and the original pronunciation. Additionally, embodiments output, by the IoT sensor set, the intonation and the original pronunciation to the two or more users in the private chat group and output, by the IoT sensor set and a client computer, the intonation and the de-content pronunciation to one or more users outside the private chat group.

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Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention relates generally to data privacy, and more particularly to the field of improving the privacy within digital communication.

(2) The term chat room, or chatroom (and sometimes chat group), is primarily used to describe any form of synchronous conferencing, occasionally even asynchronous conferencing. The term can thus mean any technology, ranging from real-time online chat and online interaction with strangers (e.g., online forums) to fully immersive graphical social environments. The primary use of a chat room is to share information via text with a group of other users. The ability to converse with multiple people in the same conversation differentiates chat rooms from instant messaging programs, which are more typically designed for one-to-one communication. The users in a particular chat room are generally connected via a shared internet or other similar connection, and chat rooms exist catering for a wide range of subjects. New technology has enabled the use of file sharing and webcams, and virtual interaction using three-dimensional avatars within a virtual world.

SUMMARY

(3) Embodiments of the present invention disclose a computer-implemented method, computer system, and computer program product for improving digital privacy and communication, the computer-implemented method comprising: responsive to identifying a private chat group has been created between two or more users, collecting, by an internet of things (IoT) sensor set, a voice for each user in the private chat group; analyzing, by an analysis module, the collected voice, an intonation associated with the collected voice and an original pronunciation associated with the collected voice; generating a de-content pronunciation based on the analysis of the collected voice, the intonation, and the original pronunciation; outputting, by the IoT sensor set, the intonation and the original pronunciation to the two or more users in the private chat group; and outputting, by the IoT sensor set and a client computer, the intonation and the de-content pronunciation to one or more users outside the private chat group.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a functional block diagram illustrating a distributed data processing environment, a program, in accordance with an embodiment of the present invention;

(2) FIG. 1B is a functional block diagram illustrating a distributed data processing environment, a program, in accordance with an embodiment of the present invention;

(3) FIG. 2 illustrates a functional block diagram and operational steps of the program, on a server computer within the distributed data processing environment of FIGS. 1, in accordance with an embodiment of the present invention;

(4) FIG. 3 illustrates a functional block diagram and operational steps of the program, on a server computer within the distributed data processing environment of FIGS. 1, in accordance with an embodiment of the present invention;

(5) FIG. 4 illustrates a functional block diagram and operational steps of the program, on a server computer within the distributed data processing environment of FIGS. 1, in accordance with an embodiment of the present invention;

(6) FIG. 5 illustrates operational steps of the program, on a server computer within the distributed data processing environment of FIGS. 1, for enhancing privacy protection of a chat group in a virtual environment, in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

(7) Embodiments of the present invention recognize that private chat groups in three-dimensional (3D) virtual rooms do not disclose private chat content to users outside the group. Typically, a floating text above a private chat group is used at most to inform external users of the current topic being discussed by the private chat group, which allows external users gauge their interest in joining the chat group. However, embodiments of the present invention recognize that using floating text to inform external users of the current topic does not create a good experience and is not a natural way of knowing the current discussion topic that does not mimic or mirror conversations in the physical world.

(8) Moreover, embodiments of the present invention recognize that before users join private chat groups, due to the opacity of these groups, users are unable determine the overall engagement level of the group, i.e., whether there are many or few utterances taking place, the engagement level of each member within the group, i.e., how many utterances each member has made, and/or understand the progress of the current topic (i.e., how long the current topic will last). Therefore, embodiments of the present invention recognize that after applying and being approved to join a private group, users may find that the group does not talk about topics efficiently, or the current topic that they are interested has ended, which results an unsatisfactory experience for users. Further, embodiments of the present invention recognize that a user's voice may be considered sensitive and/or private data because a user's voice is unique and can be utilized to access or identify sensitive and/or protected data. Embodiments of the present invention recognize that, currently in the art, to protect the privacy in metaverse, one known solution is to only transfer the voice among limited people, which leaves others unable to hear the voice at all. This current solution makes a virtual world (e.g., the metaverse) a strange place because it leaves users to witness other users silently speaking with their mouths and bodies (e.g., body language).

(9) Embodiments of the present invention improve the art of virtual communication and solve the issues stated above by presenting external perceivable features of a private chat group within a virtual world. More specifically, embodiments improve the art of virtual communication and solve the particular issues stated above by (i) inviting an attendee or a group of people to join a group in privacy mode; (ii) capturing the voice and analyzing it to get an intonation, an original pronunciation, and a de-content pronunciation; and (iii) broadcasting the intonation and original pronunciation to the attendee in the privacy mode and the intonation and de-content pronunciation to a person outside the privacy mode. Pronunciation may be the context of a voice (e.g., “it is a good day”, “how are you”, etc.) and may comprise both phrases, sentences, and any other form of context of speech by an identified voice.

(10) Implementation of embodiments of the invention may take a variety of forms, and exemplary implementation details are discussed subsequently with reference to the Figures (i.e., FIG. 1A-FIG. 5).

(11) Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

(12) A computer program product embodiment (“CPP embodiment” or “CPP”) is a term used in the present disclosure to describe any set of one, or more, storage media (also called “mediums”) collectively included in a set of one, or more, storage devices that collectively include machine

readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A “storage device” is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation, or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

(13) Computing environment **100** contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as program (component) **150**. In addition to component **150**, computing environment **100** includes, for example, computer **101**, wide area network (WAN) **102**, end user device (EUD) **103**, remote server **104**, public cloud **105**, and private cloud **106**. In this embodiment, computer **101** includes processor set **110** (including processing circuitry **120** and cache **121**), communication fabric **111**, volatile memory **112**, persistent storage **113** (including operating system **122** and component **150**, as identified above), peripheral device set **114** (including user interface (UI) device set **123**, storage **124**, and Internet of Things (IoT) sensor set **125**), and network module **115**. Remote server **104** includes remote database **130**. Public cloud **105** includes gateway **140**, cloud orchestration module **141**, host physical machine set **142**, virtual machine set **143**, and container set **144**.

(14) COMPUTER **101** may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, a virtual reality headset, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network, or querying a database, such as remote database **130**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment **100**, detailed discussion is focused on a single computer, specifically computer **101**, to keep the presentation as simple as possible. Computer **101** may be located in a cloud, even though it is not shown in a cloud in FIG. **1**. On the other hand, computer **101** is not required to be in a cloud except to any extent as may be affirmatively indicated.

(15) PROCESSOR SET **110** includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry **120** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **120** may implement multiple processor threads and/or multiple processor cores. Cache **121** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **110**. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In some computing environments, processor set **110** may be designed for working with qubits and

performing quantum computing.

(16) Computer readable program instructions are typically loaded onto computer **101** to cause a series of operational steps to be performed by processor set **110** of computer **101** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as “the inventive methods”). These computer readable program instructions are stored in various types of computer readable storage media, such as cache **121** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **110** to control and direct performance of the inventive methods. In computing environment **100**, at least some of the instructions for performing the inventive methods may be stored in component **150** in persistent storage **113**.

(17) COMMUNICATION FABRIC **111** is the signal conduction path that allows the various components of computer **101** to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

(18) VOLATILE MEMORY **112** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, volatile memory **112** is characterized by random access, but this is not required unless affirmatively indicated. In computer **101**, the volatile memory **112** is located in a single package and is internal to computer **101**, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer **101**.

(19) PERSISTENT STORAGE **113** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **101** and/or directly to persistent storage **113**. Persistent storage **113** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid-state storage devices. Operating system **122** may take several forms, such as various known proprietary operating systems or open-source Portable Operating System Interface-type operating systems that employ a kernel. The code included in component **150** typically includes at least some of the computer code involved in performing the inventive methods.

(20) PERIPHERAL DEVICE SET **114** includes the set of peripheral devices of computer **101**. Data communication connections between the peripheral devices and the other components of computer **101** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion-type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **123** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **124** may be persistent and/or volatile. In some embodiments, storage **124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **101** is required to have a large amount of storage (for example, where computer **101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a

thermometer and another sensor may be a motion detector.

(21) NETWORK MODULE **115** is the collection of computer software, hardware, and firmware that allows computer **101** to communicate with other computers through WAN **102**. Network module **115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **115** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **101** from an external computer or external storage device through a network adapter card or network interface included in network module **115**.

(22) WAN **102** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN **102** may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

(23) END USER DEVICE (EUD) **103** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **101**), and may take any of the forms discussed above in connection with computer **101**. EUD **103** typically receives helpful and useful data from the operations of computer **101**. For example, in a hypothetical case where computer **101** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **115** of computer **101** through WAN **102** to EUD **103**. In this way, EUD **103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **103** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

(24) REMOTE SERVER **104** is any computer system that serves at least some data and/or functionality to computer **101**. Remote server **104** may be controlled and used by the same entity that operates computer **101**. Remote server **104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **101**. For example, in a hypothetical case where computer **101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **101** from remote database **130** of remote server **104**.

(25) PUBLIC CLOUD **105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economics of scale. The direct and active management of the computing resources of public cloud **105** is performed by the computer hardware and/or software of cloud orchestration module **141**. The computing resources provided by public cloud **105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **142**, which is the universe of physical computers in and/or available to public cloud **105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **143** and/or containers from container set **144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **141** manages

the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **140** is the collection of computer software, hardware, and firmware that allows public cloud **105** to communicate through WAN **102**.

(26) Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as “images.” A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, central processing unit (CPU) power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

(27) PRIVATE CLOUD **106** is similar to public cloud **105**, except that the computing resources are only available for use by a single enterprise. While private cloud **106** is depicted as being in communication with WAN **102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **105** and private cloud **106** are both part of a larger hybrid cloud.

(28) Component **150** may generate and output externally perceivable features (e.g., auditory, textual, and/or physical conversation) of a private chat group in a 3D virtual chat room. Physical conversation is to be construed as any form of sign language and body language. For example, suppose there is a user represented by a virtual avatar, who is not a member of a private chat group, standing in a virtual coffee shop next to a private chat group sitting at a table. In this example, responsive to component **150** identifying the user being within a predetermined distance of the private chat group, component **150** captures the voice of each member of the private chat group, analyzes the captured voice(s) to generate an intonation, an original pronunciation, and a de-content pronunciation for each voice, and broadcasts the intonation and original pronunciation to the users in the private chat group and the intonation and de-content pronunciation to user outside (i.e., not a member of) the private chat group. In various embodiments, component **150** manipulates the intonation associated with the original pronunciation to create a secondary intonation that is broadcast with the de-content pronunciation to the one or more users outside of the chat group. For example, manipulating the pitch, dynamic, timbre, and/or duration of the users captured voice associated with original pronunciation to remove and/or distort the content in the original pronunciation to create the de-content pronunciation.

(29) In various embodiments, component **150** creates a private chat group by inviting users into a group with an active privacy mode. Responsive to two or more users joining the group and the privacy mode being activated, component **150** may capture the voices of each member of the private chat group, analyzes the captured voice(s) to generate an intonation, an original pronunciation, and a de-content pronunciation for each voice. In various embodiments, component **150** outputs the intonation and original pronunciation to the users in the private chat group and the intonation and de-content pronunciation to user outside (i.e., not a member of) the private chat group. In various embodiments, the intonation, original pronunciation, and de-content pronunciation are recorded and labeled as different data (e.g., data1, data2, data3, etc.) correspondingly for each user in the chat group.

(30) In various embodiments, component **150** monitors a predetermined area in a virtual world, identifies that the two or more users are engaging in a conversation, generates and outputs a responsive prompt to each participant in a conversation to enter the private chat group, and responsive to the two or more users accepting to enter the private chat group, creates the private chat group. In various embodiments, responsive to identifying a private chat group has been created between two or more users, component **150** collects a voice for each user in the private chat group, analyzes the collected voice, an intonation associated with the collected voice and an original pronunciation associated with the collected voice, and generates a de-content pronunciation based on the analysis of the collected voice, the intonation, and the original pronunciation. Component **150** may output the intonation and the original pronunciation to the two or more users in the private chat group and output the intonation and the de-content pronunciation to one or more users outside the private chat group. In various embodiments, responsive to receiving accessibility settings, component **150** outputs non-sensical text and/or random images as the de-content pronunciation, monitors, and tracks, in real time, at least a portion of a conversation being held within the chat group and identifies and extracts an intonation associated with each voice spoken in the chat group. Component **150** may utilize the intonation to manipulate the original pronunciation and produce the de-content pronunciation, wherein the de-content pronunciation comprises nonsensical audible sounds, and wherein manipulation comprises: replaces at least a portion of content in the original pronunciation with predetermined or random pronunciations.

(31) FIG. **1B** is a functional block diagram illustrating a distributed data processing environment, generally designated **100**, in accordance with one embodiment of the present invention. The term “distributed” as used in this specification describes a computer system that includes multiple, physically distinct devices that operate together as a single computer system. FIG. **1B** provides only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environment may be made by those skilled in the art without departing from the scope of the invention as recited by the claims. Distributed data processing environment **100** includes predetermined area **158**, user **160**, client computer **101**, and remote server **104** interconnected via WAN **102**.

(32) In the depicted embodiment, via UI device set **123**, component **150** issues and/or receives permission from a user/operator of IoT sensor set **125** to track and/or monitor user **160** and predetermined area **158** (i.e., receive opt-in response **161**), wherein predetermined area **158** is a predetermined area surrounding user **160** in a real and/or virtual world area.

(33) Component **150** may utilize IoT sensor set **125** to identify and monitor user **160** while user **160** is interacting in a virtual world. For example, component **150** identifies, via cameras, stereoscopic display, stereo sound, microphones, accelerometers, gyroscopes, and/or any other sensors known and used in the art. In various embodiments, component **150**, via monitoring module **162**, utilizes IoT sensor set **125** to monitor the actions and/or interactions of user **160** in predetermined area **158** of a virtual world. For example, monitoring module **162** identifies user **160** is engaging in a conversation with a second user based on the utilization of the microphone within IoT sensor set **125** or a chat window that can be hosted on UI device set **123**. In various embodiments, component **150** may ingest omnidirectional input data from IoT sensor set **125** to observe and identify objects and other users in predetermined area **158**.

(34) In the depicted embodiment, responsive to identifying user **160** is engaging in a conversation with one or more second users in the virtual world, component **150**, via group chat module **164**, creates a private chat group for user **160** and the one or more second users. Component **150** may create a private chat group by inviting users who are participating in a conversation with user **160** within predetermined area **158**. In various embodiments, component **150** monitors predetermined area **158** in a virtual world, wherein responsive to component **150** identifying that two or more users are engaging in a conversation, component **150** outputs, via UI device set **123**, a responsive

prompt to each participant in the conversation to join/enter into a private chat group. Component **150** may create a private chat group by inviting users into a group with an active privacy mode. Responsive to one or more users joining the private chat group with user **160** and a privacy mode being activated, component **150**, via chat collection module **166**, may capture the voices of each member of the private chat group.

(35) In the depicted embodiment, component **150** analyzes, via analysis module **168**, the captured voices, from the private chat group, to generate an intonation, an original pronunciation, and a de-content pronunciation for each voice. In various embodiments, component **150** analyzes, via analysis module **168**, the collected voices of the users in the private chat group to obtain and/or generate an intonation of the user's voice, an original pronunciation, and a de-content pronunciation. In various embodiments, component **150**, via analysis module **168**, parses the collected voice(s) to identify and obtain an intonation and an original pronunciation from the voices of the users in the private chat group and to generate a de-content pronunciation from the original pronunciation. Component **150**, via analysis module **168**, may identify and separate the intonation and original pronunciation from a captured voice based on the parsing, wherein component **150** may create a separate file for each of the intonation and original pronunciation associated with a captured voice.

(36) Component **150**, via de-content module **170**, creates the de-content pronunciation by removing identified words and terms from the original pronunciation and/or by manipulating the intonation to create nonsensical sounds, wherein the nonsensical sounds are output, via IoT sensor set **125** and/or client computer **101**, as a de-content pronunciation to users outside of the private chat group. In various embodiments, the intonation with "original pronunciation" or "de-content pronunciation" can be combined to generate a voice message that is output, via IoT sensor set **125** and/or client computer **101**, to users outside of the private chat group. In the depicted embodiment, component **150** outputs, via IoT sensor set **125** and/or client computer **101**, the original pronunciation only to members in the private chat group.

(37) In various embodiments, component **150** outputs, via IoT sensor set **125** and/or client computer **101**, the captured intonation and the captured original pronunciation associated with the users in the private chat group and outputs, via IoT sensor set **125** and/or client computer **101**, the intonation and the de-content pronunciation to users outside the private chat group. In various embodiments, responsive to receiving accessibility settings, component **150** outputs, via IoT sensor set **125** and/or client computer **101**, non-sensical text and/or random images as the de-content pronunciation. Component **150**, via group chat module **164** monitors, and tracks, in real time, at least a portion of a conversation being held within the chat group, wherein component **150**, via chat collection module **166** and analysis module **168**, identifies and extracts an intonation associated with each voice spoken in the chat group. Component **150**, via de-content module **170**, may utilize the intonation to manipulate the original pronunciation and produce the de-content pronunciation, wherein the de-content pronunciation comprises nonsensical audible sounds, and wherein manipulation comprises: replaces at least a portion of content in the original pronunciation with predetermined or random pronunciations.

(38) FIG. 2 is a functional block diagram illustrating a distributed data processing environment in accordance with one embodiment of the present invention. The term "distributed" as used in this specification describes a computer system that includes multiple, physically distinct devices that operate together as a single computer system. FIG. 2 provides only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environment may be made by those skilled in the art without departing from the scope of the invention as recited by the claims. The distributed data processing environment may include client computer **101**, and remote server **104** interconnected via WAN **102**.

(39) In various embodiments, component **150** monitors and tracks, in real time, conversation (may

also be referred to as real conversation) of users in a virtual environment. In various embodiments, during a conversation between two or more users in a virtual environment, component **150** can create a chat group by inviting the two or more users to join a chat group. In the depicted embodiment, user **202.sub.1** invites user **202.sub.2** to form chat group **201** to keep the conversation private from user **202.sub.3**. In various embodiments, chat group **201** is a private chat group. In other embodiments, chat group **201** is a chat group with the ability to turn into a private chat group based on user selected settings. For example, two users, represented by avatars in a virtual environment, are seated on a bench in park, in which one user can invite the other user to join a private chat group (i.e., chat group **201**) to maintain privacy in a public virtual space. In various embodiments, chat group **201** comprises user **202.sub.1-202.sub.N**, wherein N is any positive integer.

(40) In various embodiments, component **150** captures, analyzes, and identifies the voices used when speaking in chat group **201** by the users within chat group **201**. In the depicted embodiment, component **150** captures, analyzes **204**, and handles (e.g., identifies, isolates, manipulates, and/or retrieves) the voice of user **202.sub.1** to generate and/or obtain intonation **206**, original pronunciation **210**, and de-content pronunciation **212**. The intonation may be obtained and/or generated through known techniques in the art (e.g., modeling of intonation contours). In various embodiments, component **150** converts the captured and identified original pronunciation into text to create a text mode via natural language processing. Component **150** may utilize the text in the text mode to modify the text to create the de-content pronunciation and utilize the captured intonation to broadcast the de-content pronunciation. For example, modifying “it is a good day” to “o o o ooo oo.” In various embodiments, responsive to obtaining the intonation and pronunciation in text mode, outputting a first communication to the users in the chat group and a second communication to users outside of the chat group, wherein the first communication is the original pronunciation, and the second communication is the de-content communication. In some embodiments, the second communication may a combination of the first and second communication. A communication may be any form of communication known in the art (e.g., audible, visual, etc.).

(41) In the depicted embodiment, component **150** analyzes the captured voice associated with user **202.sub.1** to obtain intonation **206** and pronunciation **208**. Component **150** may manipulate intonation **206** and pronunciation **208** to generate and output de-content pronunciation **212**. De-content pronunciation **212** is original pronunciation **210** with at least a portion of the content in the pronunciation is replaced with predetermined or random pronunciations. In various embodiments, the content modification to the pronunciation resulting in de-content pronunciation **212** is meaningless (e.g., random sounds). For example, if original pronunciation **210** spoken by user **202.sub.1** is “it is a good day” then de-content pronunciation **212** is “OO OO O OOO.” In various embodiments, component **150**, utilizing intonation **206**, manipulates original pronunciation **210** to produce de-content pronunciation **212**.

(42) In various embodiments, a user may be an automated chatbot, a user in the traditional sense, and/or any other type of user known and understood in the art. In various embodiments, responsive to received accessibility settings, output non-sensical text and/or random images as de-content pronunciation **212**.

(43) FIG. 3 is a functional block diagram illustrating a distributed data processing environment in accordance with one embodiment of the present invention. The term “distributed” as used in this specification describes a computer system that includes multiple, physically distinct devices that operate together as a single computer system. FIG. 3 provides only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environment may be made by those skilled in the art without departing from the scope of the invention as recited by the claims. The distributed data processing environment may include client computer **101**, and

remote server **104** interconnected via WAN **102**.

(44) In various embodiments, users in chat group **201** hear intonation **206** and original pronunciation **210**. In the depicted embodiment, continuing the embodiment in FIG. 2, user **202.sub.1** and user **202.sub.2** are in chat group **201**, wherein user **202.sub.1** is conversing with user **202.sub.2**. In the depicted embodiment, user **202.sub.2** hears user **202.sub.1** say “it is a good day.”

(45) FIG. 4 is a functional block diagram illustrating a distributed data processing environment in accordance with one embodiment of the present invention. The term “distributed” as used in this specification describes a computer system that includes multiple, physically distinct devices that operate together as a single computer system. FIG. 4 provides only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environment may be made by those skilled in the art without departing from the scope of the invention as recited by the claims. The distributed data processing environment may include client computer **101**, and remote server **104** interconnected via WAN **102**.

(46) In various embodiments, outputs intonation **206** and de-content pronunciation **212** to the users outside of group **201** (e.g., user **202.sub.3**). In the depicted embodiment, continuing the embodiment in FIG. 2 and FIG. 3 above, user **202.sub.1** and user **202.sub.2** are in chat group **201**, wherein component **150** outputs intonation **206** and de-content pronunciation **212** to user **202.sub.3** so that only intonation **206** and de-content pronunciation **212** are heard by user **202.sub.3**. In various embodiments, the content produced (e.g., said) by users in chat group **201** is kept private while an audible noise is output to users outside of chat group **201**. For example, user **202.sub.3** will hear “OO OO O OOO” instead of “it is a good day.” In various embodiments, component **150** manipulates intonation **206**, removes the content from any spoken and/or communicated language during a conversation, utilizes intonation **206** to replace the content with an audible sound to produce and output de-content pronunciation **212**. In various embodiments, component **150** repeats this process for each user in chat group **201** and every time a user in chat group **201** speaks. In some embodiments, component **150** continually executes the actions above until the privacy feature is disabled.

(47) FIG. 5 illustrates operational steps of component **150**, generally designated **500**, in communication with client computer **101**, remote server **104**, private cloud **106**, EUD **103**, and/or public cloud **105**, within distributed data processing environment **100**, for enhancing privacy protection of a chat group in a virtual environment, in accordance with an embodiment of the present invention. FIG. 5 provides an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environment may be made by those skilled in the art without departing from the scope of the invention as recited by the claims.

(48) In step **502**, component **150** creates a group. In various embodiments, component **150** creates a private chat group by inviting users participating in a conversation. In various embodiments, component **150** monitors a predetermined area in a virtual world, wherein if component **150** identifies that two or more users are engaging in a conversation, then component **150** outputs a responsive prompt to each participant in the conversation to join/enter into a private chat group.

(49) In step **504**, component **150** collects a voice from the group. In various embodiments, component **150** collects the voice of each user in the private chat group in real-time while users are conversing in the private chat group.

(50) In step **506**, component **150** analyzes the collected voice. In various embodiments, component **150** analyzes the collected voices of the users in the private chat group to obtain and/or generate an intonation of the user's voice, an original pronunciation, and a de-content pronunciation. In various embodiments, component **150** parses the collected voice to identify and obtain an intonation of the users voice, an original pronunciation and to generate a de-content pronunciation from the original pronunciation. Component **150** creates the de-content pronunciation by removing identified words

and terms from the original pronunciation and/or by manipulating the intonation to create nonsensical sounds, wherein the nonsensical sounds are output as the de-content pronunciation. In various embodiments, the intonation with “original pronunciation” or “de-content pronunciation” can be combined to generate a voice message.

(51) In step **508**, component **150** outputs an intonation and original pronunciation to the chat group. In various embodiments, component **150** outputs an intonation and original pronunciation, associated with the collected voice, to the chat group.

(52) In step **510**, component **150** outputs the intonation and de-content pronunciation to a user outside of the chat group. In various embodiments, component **150** outputs an intonation and the de-content pronunciation to user(s) that are not a member of the private chat group.

(53) The programs described herein are identified based upon the application for which they are implemented in a specific embodiment of the invention. However, it should be appreciated that any particular program nomenclature herein is used merely for convenience, and thus the invention should not be limited to use solely in any specific application identified and/or implied by such nomenclature.

(54) The present invention may be a system, a method, and/or a computer program product. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

(55) Computer readable program instructions described herein may be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

(56) Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++ or the like, and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

(57) Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, may be implemented by computer readable program instructions.

(58) These computer readable program instructions may be provided to a processor of a general-purpose computer, a special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that may direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

(59) The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus, or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

(60) The flowchart and block diagrams in the Figures (i.e., FIG.) illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, a segment, or a portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, may be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

(61) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention. The terminology used herein was chosen to best explain the principles of the embodiment, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A computer-implemented method comprising: responsive to identifying a private chat group has been created between two or more users, collecting, by an internet of things (IoT) sensor set, a voice for each user in the private chat group; analyzing, by an analysis module, the collected voice, an intonation associated with the collected voice and an original pronunciation associated with the collected voice; generating a de-content pronunciation based on the analysis of the collected voice, the intonation, and the original pronunciation; outputting, by the IoT sensor set, the intonation and the original pronunciation to the two or more users in the private chat group; and outputting, by the IoT sensor set and a client computer, the intonation and the de-content pronunciation to one or more users outside the private chat group.

2. The computer-implemented method of claim 1, further comprising: responsive to receiving accessibility settings, outputting non-sensical text and/or random images as the de-content

pronunciation.

3. The computer-implemented method of claim 1, further comprising: monitoring and tracking, in real time, at least a portion of a conversation being held within the chat group; and identifying and extracting an intonation associated with each voice spoken in the chat group.

4. The computer-implemented method of claim 1, further comprising: utilizing the intonation to manipulate the original pronunciation and produce the de-content pronunciation, wherein manipulation comprises: replacing at least a portion of content in the original pronunciation with predetermined or random pronunciations.

5. The computer-implemented method of claim 4, wherein the de-content pronunciation comprises nonsensical audible sounds.

6. The computer-implemented method of claim 1, further comprising: monitoring a predetermined area in a virtual world; and identifying that the two or more users are engaging in a conversation.

7. The computer-implemented method of claim 1, further comprising: generating and outputting a responsive prompt to each participant in a conversation to enter the private chat group; and responsive to the two or more users accepting to enter the private chat group, creating the private chat group.

8. A computer system comprising: one or more computer processors; one or more computer readable storage devices; program instructions stored on the one or more computer readable storage devices for execution by at least one of the one or more computer processors, the stored program instructions comprising: responsive to identifying a private chat group has been created between two or more users, program instructions to collect, by an internet of things (IoT) sensor set, a voice for each user in the private chat group; program instructions to analyze, by an analysis module, the collected voice, an intonation associated with the collected voice and an original pronunciation associated with the collected voice; program instructions to generate a de-content pronunciation based on the analysis of the collected voice, the intonation, and the original pronunciation; program instructions to output, by the IoT sensor set, the intonation and the original pronunciation to the two or more users in the private chat group; and program instructions to output, by the IoT sensor set and a client computer, the intonation and the de-content pronunciation to one or more users outside the private chat group.

9. The computer system of claim 8, further comprising: responsive to receiving accessibility settings, program instructions to output non-sensical text and/or random images as the de-content pronunciation.

10. The computer system of claim 8, further comprising: program instructions to monitor and track, in real time, at least a portion of a conversation being held within the chat group; and program instructions to identify and extract an intonation associated with each voice spoken in the chat group.

11. The computer system of claim 8, further comprising: program instructions to utilize the intonation to manipulate the original pronunciation and produce the de-content pronunciation, wherein manipulation comprises: program instructions to replace at least a portion of content in the original pronunciation with predetermined or random pronunciations.

12. The computer system of claim 11, wherein the de-content pronunciation comprises nonsensical audible sounds.

13. The computer system of claim 8, further comprising: program instructions to monitor a predetermined area in a virtual world; and program instructions to identify that the two or more users are engaging in a conversation.

14. The computer system of claim 8, further comprising: program instructions to generate and output a responsive prompt to each participant in a conversation to enter the private chat group; and responsive to the two or more users accepting to enter the private chat group, program instructions to create the private chat group.

15. A computer program product comprising: one or more computer readable storage devices and

program instructions stored on the one or more computer readable storage devices, the stored program instructions comprising: responsive to identifying a private chat group has been created between two or more users, program instructions to collect, by an internet of things (IoT) sensor set, a voice for each user in the private chat group; program instructions to analyze, by an analysis module, the collected voice, an intonation associated with the collected voice and an original pronunciation associated with the collected voice; program instructions to generate a de-content pronunciation based on the analysis of the collected voice, the intonation, and the original pronunciation; program instructions to output, by the IoT sensor set, the intonation and the original pronunciation to the two or more users in the private chat group; and program instructions to output, by the IoT sensor set and a client computer, the intonation and the de-content pronunciation to one or more users outside the private chat group.

16. The computer program product of claim 15, further comprising: responsive to receiving accessibility settings, program instructions to output non-sensical text and/or random images as the de-content pronunciation.

17. The computer program product of claim 15, further comprising: program instructions to monitor and track, in real time, at least a portion of a conversation being held within the chat group; and program instructions to identify and extract an intonation associated with each voice spoken in the chat group.

18. The computer program product of claim 15, further comprising: program instructions to utilize the intonation to manipulate the original pronunciation and produce the de-content pronunciation, wherein the de-content pronunciation comprises nonsensical audible sounds, and wherein manipulation comprises: program instructions to replace at least a portion of content in the original pronunciation with predetermined or random pronunciations.

19. The computer program product of claim 15, further comprising: program instructions to monitor a predetermined area in a virtual world; and program instructions to identify that the two or more users are engaging in a conversation.

20. The computer program product of claim 15, further comprising: program instructions to generate and output a responsive prompt to each participant in a conversation to enter the private chat group; and responsive to the two or more users accepting to enter the private chat group, program instructions to create the private chat group.
